

Ad Click Analysis

Problem Statement

Given purchased user IDs, ad click logs (IP, ad text), and a mapping userId->IP, return each ad's total clicks and clicks that led to purchases.

Variants

- Multiple IPs per user; deduplicate repeated clicks

- Only

How to Leverage the Solution

Convert user->IP to IP->user. One pass to count totals and purchase-clicks; add optional (ip,ad) dedup set.

Java Solution

Below is a concise reference implementation. Full comments omitted for brevity.

```
import java.util.*;

public final class AdClickAnalytics {
    public static final class AdClick {
        public final String ip; public final String adText;
        public AdClick(String ip, String adText) { this.ip = ip; this.adText = adText; }
    }

    public static Map<String, int[]> analyze(Set<String> purchasedUserIds,
                                           List<AdClick> clicks,
                                           Map<String, String> userIdToIp) {
        Map<String, String> ipToUser = new HashMap<>();
        for (Map.Entry<String, String> e : userIdToIp.entrySet()) {
            if (e.getValue() != null) ipToUser.put(e.getValue(), e.getKey());
        }
        Map<String, int[]> result = new HashMap<>();
        for (AdClick c : clicks) {
            String ad = c.adText == null ? "" : c.adText.trim();
            int[] counters = result.computeIfAbsent(ad, k -> new int[2]);
            counters[0]++;
            String user = ipToUser.get(c.ip);
            if (user != null && purchasedUserIds.contains(user)) counters[1]++;
        }
        return result;
    }

    public static Map<String, int[]> analyzeDedup(Set<String> purchasedUserIds,
                                                  List<AdClick> clicks,
                                                  Map<String, String> userIdToIp) {
        Map<String, String> ipToUser = new HashMap<>();
        for (Map.Entry<String, String> e : userIdToIp.entrySet()) {
            if (e.getValue() != null) ipToUser.put(e.getValue(), e.getKey());
        }
        Map<String, int[]> result = new HashMap<>();
        Set<String> seenIpAd = new HashSet<>();
        for (AdClick c : clicks) {
            String ad = c.adText == null ? "" : c.adText.trim();
            String key = c.ip + "\u0000" + ad;
            if (!seenIpAd.add(key)) continue;
            int[] counters = result.computeIfAbsent(ad, k -> new int[2]);
        }
    }
}
```

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```
    counters[0]++;  
    String user = ipToUser.get(c.ip);  
    if (user != null && purchasedUserIds.contains(user)) counters[1]++;  
}  
return result;  
}  
}
```